A065-v0.3.8 | Source-Native Vorticity Core Closure

Blind-compatible numerical preparation - NOT certified PKLSA physics

Scope and classification

These fields are GENERATED from original parametric trefoil centreline geometry using an ASSUMED Gaussian transverse smoothing width. They are 3D arrays, but they are not measurements or evolved Euler/PKLSA fields. Gamma and enstrophy inversion is conditional on the Gaussian cross-sectional hypothesis.

Methods

Omega(x) = Gamma * integral over closed C of G_a(x-X(s)) dX(s), with G_a(R)=exp(-R^2/a^2)/(pi^(3/2) a^3). Compute divergence-free periodic FFT projection, velocity and pressure-Poisson diagnostics. For each local normal plane, compute Gamma = integral (omega dot tangent) dA and Z = integral (omega dot tangent)^2 dA. Gaussian hypothesis implies a = |Gamma|/sqrt(2 pi Z). A different model such as Rankine gives a different width.

Numerical gates

Gate	Status	Scope
G28	PASS	3D field / spectral constraints
G29	FAIL	All transverse stations admissible?
G30	PASS	At least one admissible inference per source
G31	FAIL	No-fit withheld velocity and pressure predictions
G32	UNRESOLVED	Authentic independently qualified PKLSA

Measured transverse profiles

FOURIER_A - a_seed = 0.500; periodic N=48^3; div(u) RMS=2.8e-18; Poisson residual=2e-13; projected source correction=0.023%.

Station	Gamma	Z_axial	a(Gamma,Z)	Admitted	Vel error	Pressure error
0.00	1.0661	0.5531	0.5719	YES	4.9%	18.3%
0.25	0.7278	0.7205	0.3421	NO	21.4%	53.5%
0.50	-0.0026	1.2134	0.0009	NO	100.5%	100.0%
0.75	0.7278	0.7205	0.3421	NO	21.4%	53.5%

TORUS_B - a_seed = 0.650; periodic N=48^3; div(u) RMS=2.2e-18; Poisson residual=1.5e-18; projected source correction=0.034%.

Station	Gamma	Z_axial	a(Gamma,Z)	Admitted	Vel error	Pressure error
0.00	1.0992	0.3680	0.7228	YES	3.6%	22.6%
0.25	1.0701	0.3622	0.7094	YES	2.7%	37.2%
0.50	1.2963	0.4075	0.8101	NO	7.1%	98.8%
0.75	1.0701	0.3622	0.7094	YES	2.7%	37.2%

Scientific interpretation / fail-closed limits

Some planes intersect neighbouring strands or contain strong non-axial vorticity, leading to cancellations. These are REJECTED before inference. G31 is a synthetic surrogate-prediction failure; the FFT-heldout velocity and

pressure derive from the same generated source and are NOT independently measured. The nonzero steady Euler residual shows the generated field is not established as an Euler steady state; no time evolution was solved.

For a physical claim the upstream PKLSA runner must export original u(x,y,z,t) or omega(x,y,z,t), a specified domain, boundary conditions, a transverse-core sampling region, circulation and enstrophy convergence, and a separate independently computed heldout u/p field with source hashes. The geometry-only E010/E011 outputs do not fulfil this requirement.